PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview 1
2. Background & Business Objective 2
3. Product Purpose 5
4. System Functionalities 5
5. User Stories & Use Cases 8
6. Features Included (Scope Definition) 10
7. Features Not Included (Scope Control) 10
8. Assumptions & Constraints 10
9. Risks & Questions Throughout Development 11

Project Name: BizAssist AI

Version: 1.0

1. Project Overview

- **Problem Statement:**

  Small and medium enterprises (SMEs) rely heavily on platforms like WhatsApp, Instagram, and manual spreadsheets to manage customer interactions, orders, and scheduling. These workflows are:

  - Unstructured and fragmented
  - Highly dependent on human memory
  - Prone to delayed responses and missed opportunities

For example:

  - Customers say "same as last time" → no structured record
  - Orders are incomplete → missing address, deadline
  - Messages go unanswered → leads to lost sales

- **Target Domain:** AI-powered workflow automation for **SME customer communication and operations management**

- **Proposed Solution Summary:** BizAssist AI is an **AI-powered business messaging dashboard** that transforms messy conversations into structured workflows.

  It uses an AI reasoning engine to:

  - Understand unstructured messages (Manglish, BM, English)
  - Extract orders, complaints, and requests
  - Detect risks (e.g., cancellation intent, delays)
  - Suggest actionable responses

* Instead of manually tracking chats, SMEs can **operate like structured enterprises**.

2. Background & Business Objective

- **Background of the Problem:**

  In Malaysia and globally, SMEs:

  - Use **WhatsApp as primary CRM**
  - Lack proper systems (ERP/CRM too expensive)
  - Manage orders manually via chat + memory

  This results in:

  - Slow response times (5–30 minutes average)
  - Lost conversions due to delayed replies
  - Poor customer experience

- **Importance of Solving This Issue:**

  Improving response and workflow efficiency can:

  - Increase conversion rates by **20–40%**
  - Reduce missed orders significantly
  - Improve customer retention

- **Strategic Fit / Impact:**

  This solution aligns with:

  - AI-driven productivity tools
  - SME digital transformation initiatives
  - AI-assisted workflow automation (GLM core requirement)

3. Product Purpose

3.1. Main Goal of the System: To act as an **AI workflow assistant** that helps SMEs manage customer interactions, orders, and decisions in real-time.

3.2. Intended Users (Target Audience):

- SME business owners
- Customer service agents
- Online sellers (WhatsApp/Instagram-based)
- Small operations teams

4. System Functionalities

4.1. Description: BizAssist AI acts as a **central dashboard** where:

- Conversations are displayed like a chat system
- AI continuously analyzes incoming messages
- Business actions (orders, meetings, issues) are automatically extracted

4.2. Key Functionalities:

4.2.1 **AI Conversation Analysis**

- Detect intent (Order / Complaint / Inquiry / Meeting)
- Extract structured data:
  - Items
  - Quantity
  - Address
  - Deadline

- Identify missing fields

4.2.2 Smart Reply Suggestions

- AI generates professional replies
- Agent can:
    - Use directly
    - Edit before sending
    - Regenerate

4.2.3 Workflow Automation

Convert chats into:

- Orders → Kanban board
- Meetings → Calendar
- Issues → AI flags

4.2.4 Risk Detection System

AI flags:

- Unresponded messages (>15 min)
- Cancellation intent
- Missing order details
- Ambiguous requests

4.2.5 AI Assistant (Agent Copilot)

Agents can ask:

- “What is missing from this order?”
- “Is this customer high risk?”
- “Is Friday available for meeting?”

4.2.6 Analytics Dashboard

- Messages handled per day
- Orders and revenue trends

- AI performance metrics
- Agent productivity

4.3. AI Model & Prompt Design

4.3.1 Model Selection

- **Z.AI GLM**
- **Chosen because:**
    - **Strong reasoning capability**
    - **Multilingual understanding (BM, English, Manglish)**
    - **Structured output (JSON)**

4.3.2 Prompting Strategy

- Multi-step agentic prompting:
    1. Understand intent
    2. Extract structured data
    3. Identify missing info
    4. Generate reply

4.3.3 Context & Input Handling

- Input: Full conversation thread
- If too long:
    - Summarization + chunking
- Includes:
    - Past orders
    - Customer history
    - Meeting schedules

4.3.4 Fallback & Failure Behavior

- If AI output invalid:
  - Retry once
  - Show fallback suggestion
- Agent can:
  - Override AI
  - Escalate case

5. User Stories & Use Cases

- **User Stories:**
  - As an SME owner, I want AI to summarize customer messages so I don't miss important details
  - As an agent, I want reply suggestions so I can respond faster
  - As a business, I want chats converted into orders automatically

- **Use Cases (Main Interactions):**

1. Chat-to-Workflow Flow

Users receive customer messages through the dashboard. BizAssist AI analyzes the conversation, identifies the customer's intent, extracts important details such as product, quantity, address, deadline, and issue type, then converts the message into a structured workflow such as an order, complaint, inquiry, or meeting request.

2. AI Reply Assistance Flow

The agent selects a customer conversation and clicks **"Ask AI"**. BizAssist AI reviews the chat history, detects missing information or risks, and generates a professional suggested reply. The agent can use, edit, regenerate, or reject the suggestion before sending it to the customer.

3. Order Creation Flow

When a customer places an order through chat, the AI extracts the order details and checks whether any important fields are missing. If the order is complete, the agent can confirm it and create an order card in the Kanban board. If details are missing, the AI suggests what the agent should ask next.

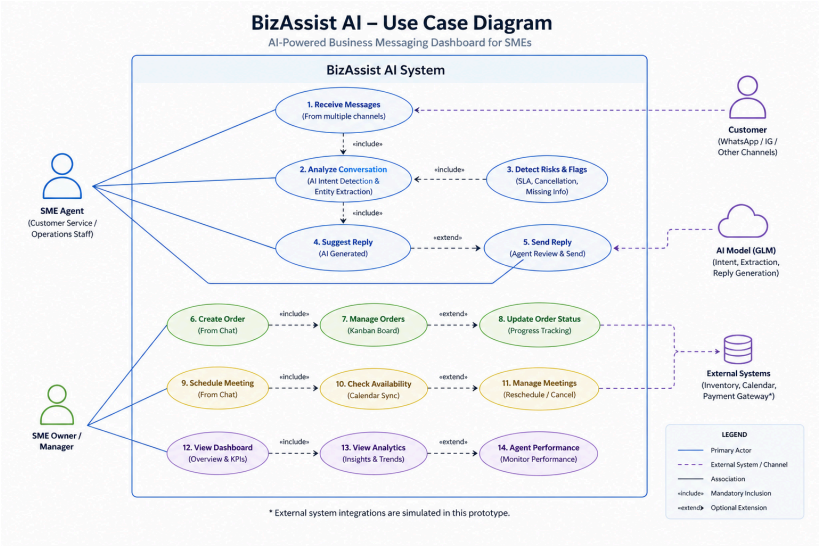
4. Risk Flagging Flow

BizAssist AI continuously monitors customer conversations for risks such as unanswered messages, cancellation intent, complaints, unclear order references, or urgent deadlines. These issues are shown as AI flags so agents can prioritize high-risk conversations first.

5. Meeting Scheduling Flow

When a customer requests a meeting, BizAssist AI identifies the proposed date and time, checks the existing meeting schedule, and warns the agent if there is a conflict. The agent can then confirm, reschedule, or escalate the meeting request.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- AI chat analysis system
- Smart reply generation
- Order management (Kanban)
- Meeting scheduling
- Customer profiles
- Product catalog
- Analytics dashboard

7. Features Not Included (Scope Control)

- Full backend ERP integration
- Payment gateway
- Multi-organization real-time sync
- Production-grade deployment

8. Assumptions & Constraints

- **LLM Cost Constraint:**
  Estimated: **$0.01–$0.03 per conversation analysis**
  Optimization:
  - Cache repeated prompts
  - Limit token usage

- **Technical Constraints:** The current system is designed as a **frontend-focused prototype (React + Vite)** with simulated backend logic and limited real-time integrations. AI processing is dependent on external API calls (e.g., GLM/LLM), which introduces latency (~1–2 seconds) and requires internet connectivity. Data persistence is handled via **localStorage**, meaning it is not scalable or suitable for production environments. Integration with external systems such as WhatsApp

Business API, payment gateways, and inventory databases is **mocked or simulated** for demonstration purposes

- **Performance Constraints:**
  - AI response delay ~1–2 seconds (simulated)
  - Large conversations require summarization

- **User Input:** The system relies on human-generated customer conversations as its primary input source. AI analysis is triggered only when new messages are received or when the agent explicitly requests assistance. While the model can interpret unstructured inputs (e.g., Bahasa Malaysia, mixed-language messages, voice notes, and images), it **does not operate autonomously** and requires human validation before any action (e.g., order confirmation, customer reply) is finalized. This ensures accuracy and prevents incorrect business decisions. Additionally, incomplete or ambiguous inputs (e.g., "same as last time") will require the agent to manually clarify missing details before proceeding.

9. Risks & Questions Throughout Development

1. AI Accuracy

- Risk: Incorrect extraction of order details
- Mitigation: Human confirmation required

2. Over-Reliance on AI

- Risk: Agents blindly trust AI
- Mitigation: Highlight confidence scores

3. Data Privacy

- Risk: Customer data exposure
- Mitigation: Local storage for demo